# A1-21MAB301T- PROBABILITY AND STATISTICS - LLT-1 (MCQ QUIZ)

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* Indicates required question

Choose the Correct Answer

The normal distribution $N(\mu, \sigma^2)$ is symmetric about: *

(A) $x = 0$

(B) $x = \sigma$

(C) $x = \mu$

(D) $x = \mu + \sigma$

- Option A
- Option B
- Option C
- Option D

If, on an average, 9 ships out of 10 arrive safely to a port, calculate the mean and standard deviation of the number of ships returning safely out of 150 ships. *

(A) Mean = 135, S.D = 3.674
(B) Mean = 153, S.D = 2.616
(C) Mean = 0, S.D = 1.112
(D) Mean = 0.9, S.D = 0.150

- Option A
- Option B
- Option C
- Option D

For a continuous random variable X, P(X = a) equals: *

(A) f(a)

(B) F(a)

(C) 1

(D) 0

- Option A
- Option B
- Option C
- Option D

The c-chart is used to monitor: *

(A) Proportion of defectives

(B) Number of defects per unit

(C) Process mean

(D) Process range

- Option A
- Option B
- Option C
- Option D

In a one-way ANOVA, how many independent variables are there? *

(A) more than one

(B) three

(C) two

(D) one

- Option A
- Option B
- Option C
- Option D

Spearman's rank correlation is used when: *

(A) Data is quantitative and normally distributed

(B) Data is qualitative or ordinal

(C) n > 30

(D) Variances are equal

- Option A
- Option B
- Option C
- Option D

Two events A and B are mutually exclusive if: *

(A) $P(A \cap B) = P(A) \cdot P(B)$

(B) $P(A \cap B) = 0$

(C) $P(A \cup B) = 1$

(D) $P(A) = P(B)$

- Option A
- Option B
- Option C
- Option D

If X follows a uniform distribution U(3, 9),what is P(4≤X≤6)? *

(A) 1/2

(B) 1/3

(C) 1/4

(D) 1/6

- Option A
- Option B
- Option C
- Option D

In a two-sample t-test for difference of means, the degrees of freedom is: *

(A) $n_1 + n_2$

(B) $n_1 + n_2 - 2$

(C) $n_1 + n_2 + 2$

(D) $n_1 \times n_2$

- Option A
- Option B
- Option C
- Option D

The UCL for p-chart is: *

(A) $\bar{p} + 3\sqrt{(\bar{p}\bar{q}/n)}$

(B) $\bar{p} - 3\sqrt{(\bar{p}\bar{q}/n)}$

(C) $\bar{p} + 3\sqrt{(\bar{p}/n)}$

(D) $\bar{p} - 3\sqrt{(\bar{p}/n)}$

- Option A
- Option B
- Option C
- Option D

If r = 0, then the variables are: *

(A) Perfectly positively correlated

(B) Perfectly negatively correlated

(C) Uncorrelated (no linear relationship)

(D) Identical

- Option A
- Option B
- Option C
- Option D

Any subset of the sample space  is defined as: *

(A) Infinite sample space

(B) Finite sample space

(C) Trial

(D) Event

- Option A
- Option B
- Option C
- Option D

For an m×n contingency table, the number of degrees of freedom equals *

(A) mn

(B) m+n

(C) (m+1)(n−1)

(D) (m−1)(n−1)

- Option A
- Option B
- Option C
- Option D

The level of significance α is the probability of committing: *

(A) Type II error

(B) No error

(C) Type I error

(D) Both errors

- Option A
- Option B
- Option C
- Option D

Poisson distribution is a limiting case of *

(A) Normal distribution

(B) Uniform distribution

(C) Geometric distribution

(D) Binomial distribution

- Option A
- Option B
- Option C
- Option D

Analysis of variance is a statistical method of comparing the several populations of *

(A) Means
(B) Variances
(C) Exponential
(D) Normals

- Option A
- Option B
- Option C
- Option D

A bag contains 5 red balls and 3 blue balls. A ball is drawn at random. What is the probability of drawing a red ball? *

(A) 3/8

(B) 1/2

(C) 5/8

(D) 2/3

- Option A
- Option B
- Option C
- Option D

An F-test is conducted with two samples of sizes 8 and 10. The calculated F-value is 4.5, and the critical F-value at a 5% significance level is 3.2. What is the conclusion? *

(A) Accept the null hypothesis

(B) Reject the null hypothesis

(C) Increase the sample size

(D) Cannot determine

- Option A
- Option B
- Option C
- Option D

The centerline in a control chart represents *

(A) The specification limit

(B) The process mean

(C) The number of defects

(D) The maximum variation allowed

- Option A
- Option B
- Option C
- Option D

The difference between p-chart and np-chart is: *

(A) p-chart uses proportion; np-chart uses count of defectives

(B) p-chart is for variable data; np-chart is for attribute data

(C) np-chart needs larger samples

(D) They are identical

- Option A
- Option B
- Option C
- Option D

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